

No. of Functional Features Included in the Solution

Date	30 June 2026
Team ID	
Project Name	AI and Machine Learning Project
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Registration/Input Form	Collects applicant details such as age, income, credit score, employment status, and other required information.	Frontend (HTML/CSS)	Done	High
2	Data Preprocessing	Cleans and preprocesses input data using label encoding and feature scaling before prediction.	Backend (Python)	Done	High
3	Machine Learning Prediction	Predicts whether the credit card application will be approved or rejected using the trained Random Forest model.	ML Model	Done	High
4	Prediction Result Display	Displays the approval/rejection result with confidence to the user.	Flask + HTML	Done	High
5	Model Loading	Loads the saved machine learning model (model.pkl) and	Backend	Done	Medium

		preprocessing files using Joblib.			
6	Web Application Deployment	Deploys the application online using PythonAnywhere for public access.	Deployment	Done	Medium
7	GitHub Version Control	Maintains project source code and version history using GitHub.	Version Control	Done	Medium
8	Error Handling & Input Validation	Validates user inputs and handles prediction errors gracefully.	Backend	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5

Additional / Nice-to-Have Features	3
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Applicant input form, Prediction result page
2	Backend / Logic	4	Flask routing, Data preprocessing, Random Forest prediction, Error handling
3	Database / Storage	1	CSV dataset and saved ML model (model.pkl)
4	API / Integration	1	Flask application integration with machine learning model
5	Security / Authentication	0	Not implemented in the current version